Database Systems and Cloud Computing

AIN-3003, Bahcesehir University, Artificial Intelligence Engineering Department

Homework 1 Report

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1. Approach

Each question is answered with a single SQL statement that is executed from a Python script. The script follows the pattern used in class: it connects to MySQL with mysql.connector, opens a cursor, sends the statement with my_cursor.execute(), then reads the returned rows in a loop and prints them with the formatting given in the assignment sheet. The statements below are the same ones that appear in hw1_queries.sql.

2. Questions and results

Question 1

Find the city of the employees in the IT department.

SQL statement

```
SELECT e.last_name, d.department_name, l.city
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id
WHERE d.department_name = 'IT';
```

Explanation

The last name is stored in employees, the department name in departments and the city in locations. The tables are linked through the foreign keys employees.department_id and departments.location_id. The filter uses an exact match on 'IT', because the table also contains departments such as 'IT Support' and 'IT Helpdesk'.

Output

```
Last Name  Department city
Hunold                                    IT                                        Southlake
Ernst                                     IT                                        Southlake
Austin                                    IT                                        Southlake
Pataballa                                 IT                                        Southlake
Lorentz                                   IT                                        Southlake
```

Question 2

Find the number of employees in each city and order in descending order.

SQL statement

```
SELECT l.city, COUNT(*) AS number_of_employees
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id
GROUP BY l.city
ORDER BY number_of_employees DESC, l.city;
```

Explanation

The same three table join is grouped by city and COUNT(*) counts the employees in each group. ORDER BY ... DESC places the largest city first, and the second sort key (city name) keeps the order stable when two cities have the same count.

Output

City	Number of Employees
South San Francisco	45
OX9 9ZB	34
Seattle	18
Southlake	5
Toronto	2
London	1
Munich	1

Question 3

Find the number of employees in each job title and order in descending order.

SQL statement

```
SELECT j.job_title, COUNT(*) AS number_of_employees
FROM employees e
JOIN jobs j ON e.job_id = j.job_id
GROUP BY j.job_title
ORDER BY number_of_employees DESC, j.job_title;
```

Explanation

The job title is stored in jobs, while employees only holds job_id, so only these two tables are joined. Employees are counted per title and sorted in descending order. The counts add up to 107, the number of rows in employees.

Output

```
Job Title                 Number employees
Sales Representative      30
Shipping Clerk            20
Stock Clerk               20
Accountant                5
Programmer                5
Purchasing Clerk          5
Sales Manager             5
Stock Manager             5
Administration Vice President 2
Accounting Manager        1
Administration Assistant 1
Finance Manager           1
Human Resources Representative 1
Marketing Manager         1
Marketing Representative 1
President                 1
Public Accountant         1
Public Relations Representative 1
Purchasing Manager        1
```

Question 4

Find the employees in UK and CANADA only.

SQL statement

```
SELECT CONCAT(e.first_name, ' ', e.last_name) AS full_name, l.city, c.country_name
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id
JOIN countries c ON l.country_id = c.country_id
WHERE c.country_name IN ('United Kingdom', 'Canada');
```

Explanation

The country name is stored in countries, so the join chain is employees, departments, locations and countries. CONCAT builds the full name, and IN filters the two countries at once.

Output

```
first name + Last Name           City country
Michael Hartstein                Toronto Canada
Pat Fay                          Toronto Canada
Susan Mavris                     London United Kingdom
```

Question 5

Find the average salary in every city.

SQL statement

```
SELECT l.city, ROUND(AVG(e.salary), 2) AS average_salary
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id
GROUP BY l.city
ORDER BY l.city;
```

Explanation

The same three table join is grouped by city, and AVG(e.salary) is calculated for each group. ROUND(..., 2) limits the result to two decimals, and the cities are listed alphabetically, as in the sample output.

Output

City	Average Salary
London	6500.00
Munich	10000.00
OX9 9ZB	8955.88
Seattle	8844.44
South San Francisco	3475.56
Southlake	5760.00
Toronto	9500.00

3. Notes on the data

- **Oxford location.** In the provided dataset the row location_id = 2500 is shifted because of a comma inside the street address, so its city value is OX9 9ZB and its country_id is Ox. The data was not modified. This is why the city appears as OX9 9ZB in Questions 2 and 5, and why the Oxford employees do not appear in Question 4, since Ox does not exist in the countries table. The sample outputs in the assignment sheet show the same behaviour.
- **Employee without a department.** One employee (Kimberely Grant, id 178) is stored with department_id = 0, which does not match any department. The inner joins therefore return 106 employees in Questions 1, 2 and 5, while Question 3, which does not use departments, counts all 107.
- **Sort order.** The sample outputs of Questions 2 and 3 are not in descending order, while the question text asks for descending order. The queries follow the question text.